Navigating the Complexities of Child Online Safety

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Introduction

Currently, child online safety focuses on monitoring, filtering, and overall controlling a child's online experience. This approach is very limited in what it can achieve and has significant drawbacks. In particular, it raises concerns such as a potential lack of freedom, poor digital skills, deterioration of parent-child relationships, and potential data privacy issues.

Proposed Solution

The main challenge with effective child online safety solutions is finding a balance between protecting children from potential threats such as cyberbullying, phishing, and online predation (Figure 1) while still allowing them the freedom to benefit from the internet. This makes the issue socio-technical by nature; understanding this allows us to analyze current tools and help us create better safety tools.

To address the complex issue a new tool must be made that can offer personalized guidance while not being overly restrictive. This means an effective solution will be AI driven.

Figure 1: Child online safety involves many threats such as cyberbullying, phishing, and online predation (e.g. children being blackmailed with inappropriate AI generated images).

Research Question

"How can child online safety tools be improved to address the new, growing, and complex online threats?"

Research

The project has three key research areas: child online safety, AI for parental control, and the personalization of safety solutions. The first step of research was to analyze trends in these categories. To do this papers were systematically collected (Figure 2) from the ACM Digital Library.

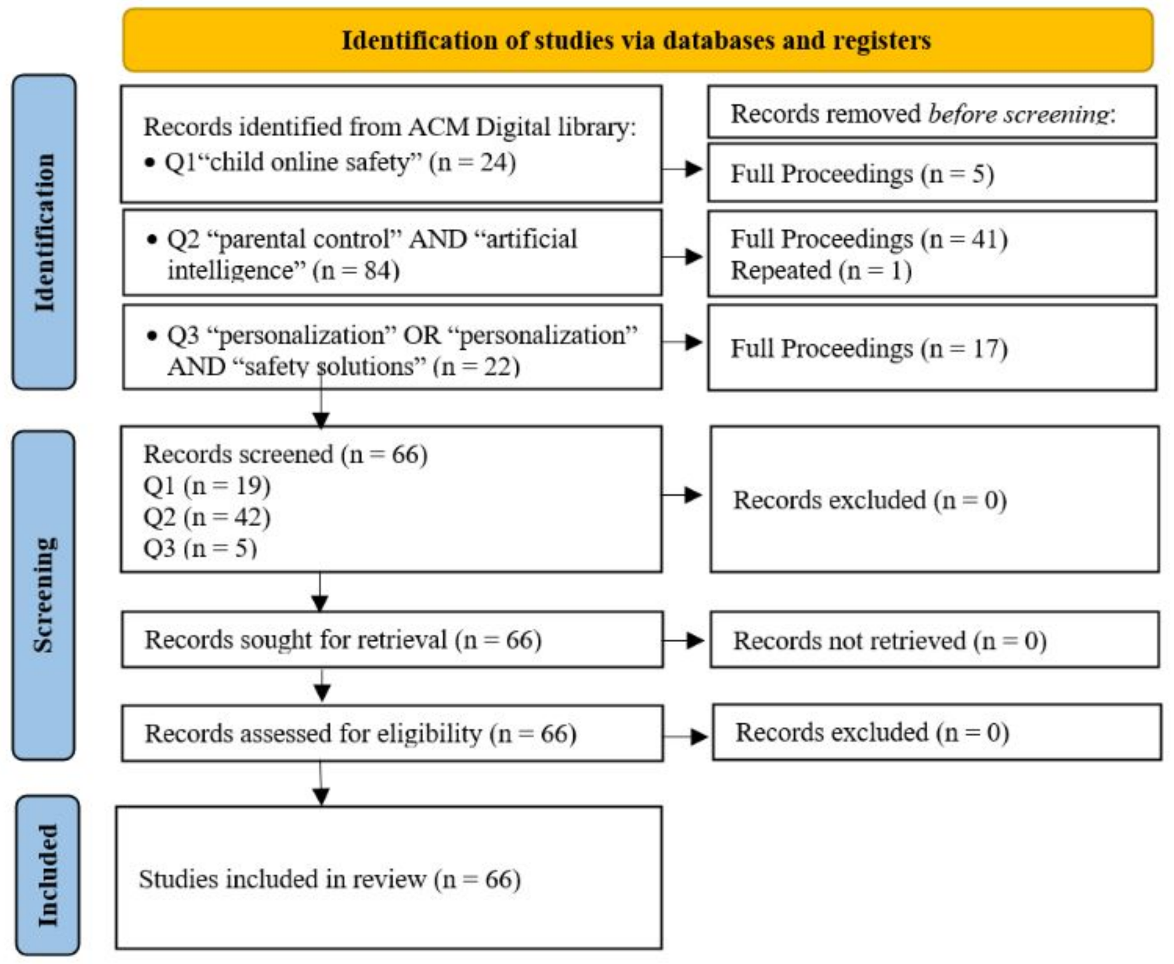
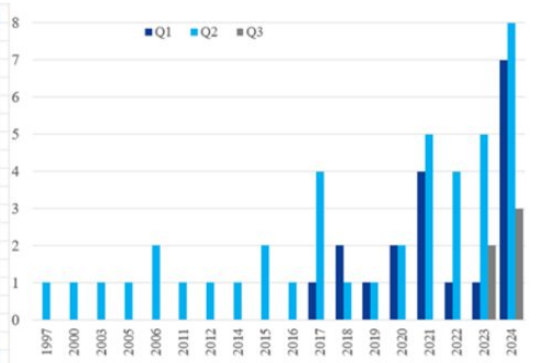


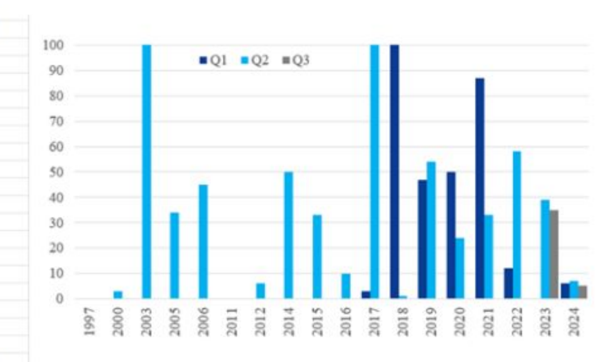
Figure 2: PRISMA guideline applied to the collection of papers from the ACM Digital Library. N is the number of papers at each step.

The papers were then analyzed qualitatively and quantitatively. Qualitative analysis is based in the content of the papers themselves while quantitative analysis is performed on properties such as date of publication (Figures 3, Figure 4).

Year	Q1	Q2	Q3	Total
1997		1		1
2000		1		1
2003		1		1
2005		1		1
2006		2		2
2011		1		1
2012		1		1
2014		1		1
2015		2		2
2016		1		1
2017	1	4		5
2018	2	1		3
2019	1	1		2
2020	2	2		4
2021	4	5		9
2022	1	4		5
2023	1	5	2	8
2024	7	8	3	18
Total	19	42	5	66



Year	Q1	Q2	Q3	Total
1997		0		0
2000		3		3
2003		4458		4458
2005		34		34
2006		45		45
2011		0		0
2012		6		6
2014		50		50
2015		33		33
2016		10		10
2017	3	218		221
2018	169	1		170
2019	47	54		101
2020	50	24		74
2021	87	33		120
2022	12	58		70
2023	0	39	35	74
2024	6	7	5	18
Total	374	5073	40	5487



Q1 is sources found by ("child online safety"), Q2 by ("parental control" AND "artificial intelligence"), and Q3 by ("personalization" OR "personalization" AND "safety solutions")

Figure 3: Number of publications per year for each of the three search queries in the ACM Digital Library.

Figure 4: The annual sum of citations for the publications retrieved through the three search queries (Q1, Q2, and Q3).

Additional research was also conducted on tools and incidents.

Methodology

The goal of the methodology is to have a systematic and less biased approach to collecting data. This was mainly achieved through mixed method collection and PRISMA.

Mixed method collection is broken into quantitative and qualitative data collection with analysis after each individual part (Figure 5).

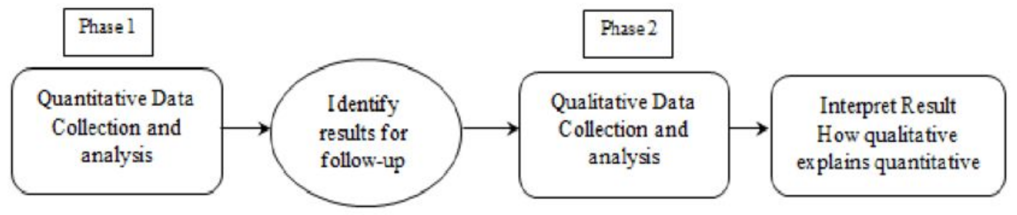


Figure 5: Flowchart for the phases of mixed method data collection.

PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) is a search guideline that breaks data collection into a sequence of steps (Figure 6).

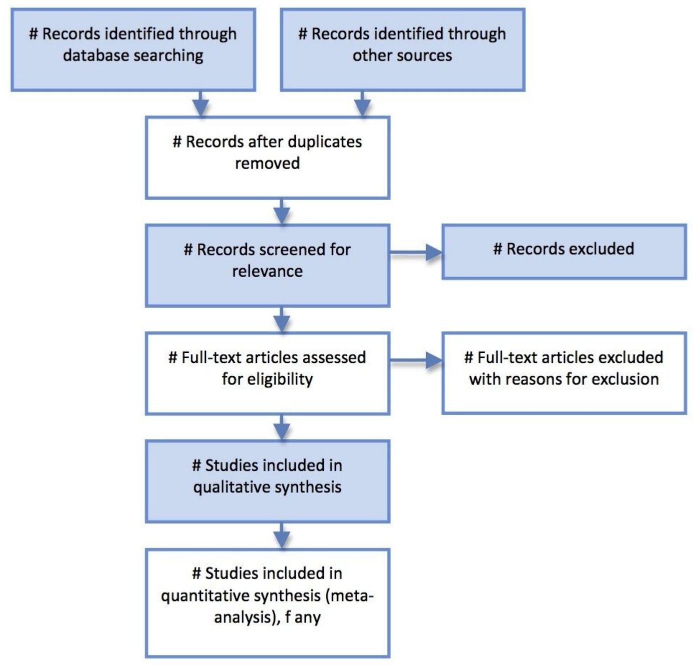


Figure 6: Flowchart for the various steps of PRISMA research guideline.

Results

Quantitatively, the numbers of publications on the three categories showed noticeable growth over time. The number of citations shows moderate interest, especially in more recent years, in child online safety; a general interest in AI for parental control; and very low interest in personalized solutions.

Qualitatively, the research shows a shift from tools focused on restriction to those focused on guidance, AI has been evaluated to be inconsistent for addressing delicate situations such as online grooming, and ethical concerns around AI models such as models being too Western-centric also arose.

References

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